Formula 1 Podium Prediction

Milestone 1 — Data engineering, leakage-safe modelling and explainability

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Summary

- **Task:** for every driver in every race, predict a podium finish (*positionOrder* ≤ 3) using only information available after qualifying and before the start.
- **Data:** the 14-table Ergast / Kaggle dataset (1950–2024), cleaned into one leakage-safe row per (race, driver). Raw files are never modified; every cleaning rule is logged and asserted.
- **Model:** logistic regression reaches **test PR-AUC 0.698** (ROC-AUC 0.934, F1 0.690) on the unseen 2022–24 seasons, a new regulation era. Ranking by starting grid scores 0.647. A random forest, gradient boosting and a shallow neural network are statistically indistinguishable from it (race-bootstrap intervals), so the simplest model is kept.
- **What drives predictions:** the starting position. Removing grid and qualifying together costs 0.114 validation PR-AUC (95% CI excludes 0); standings, form and pace each add little on top of it.
- **Data questions:** front-row conversion rankings change after 2014 (Spearman ρ = 0.21); there is no home-race advantage once grid position is controlled for (odds ratio 0.95, 95% CI 0.80–1.13); mechanical-retirement rates roughly halve from 2014–21 to 2022–24.

1. Task and data

The unit of analysis is one driver entering one race, and the target is a podium finish. The prediction is made at a fixed **cutoff**: after the starting grid is known and before the race starts. Results, points, status, laps, pit stops, lap times and post-race standings of the current race are only known after it, so using them as features would be **data leakage**. Every feature in this project is either known at the cutoff (qualifying, grid, sprint result) or computed only from earlier races.

The dataset is relational. Most questions need multi-hop joins, e.g. results joined with races, circuits, drivers, status and qualifying. All work is in one run-all notebook (*F1_Podium_Prediction.ipynb*), which runs top to bottom without manual steps and produces every table and figure in this report. A fresh re-run reproduces every output table exactly.

2. Data audit and cleaning

2.1 Audit and EDA before cleaning

- **Sentinels.** Missing values are stored as the literal text \N (Figure 1). They are concentrated in *structural* columns (session times recorded only from 2021, fastest-lap data from 2004) and *outcome* columns (*results.time* 71% missing for drivers not on the lead lap, *position* 41% for non-classified drivers). Only *q2/q3* are missing for a pre-race reason: the driver was knocked out.
- **Keys.** All primary keys are unique and non-null and every foreign key resolves. The one violation is the required grain: 176 result rows form 85 duplicated (race, driver) pairs. These are 1950s shared drives, where drivers swapped cars mid-race (Figure 2).

- **Coverage drift.** Qualifying rows exist from 1994 but cover only 6–59% of entries in 1996–2002. They are complete from 2003 (Figure 3). Lap times start in 1996, pit stops in 2011 and sprints in 2021.
- **Formats and categories.** Durations appear as *m:ss.sss*, *mm:ss.sss* and *h:mm:ss.sss*, and *results.time* mixes absolute times with gaps (22 formats). There are spelling issues ('Argentinian ' with a trailing space, 'USA' vs 'United States') and one HTML entity.

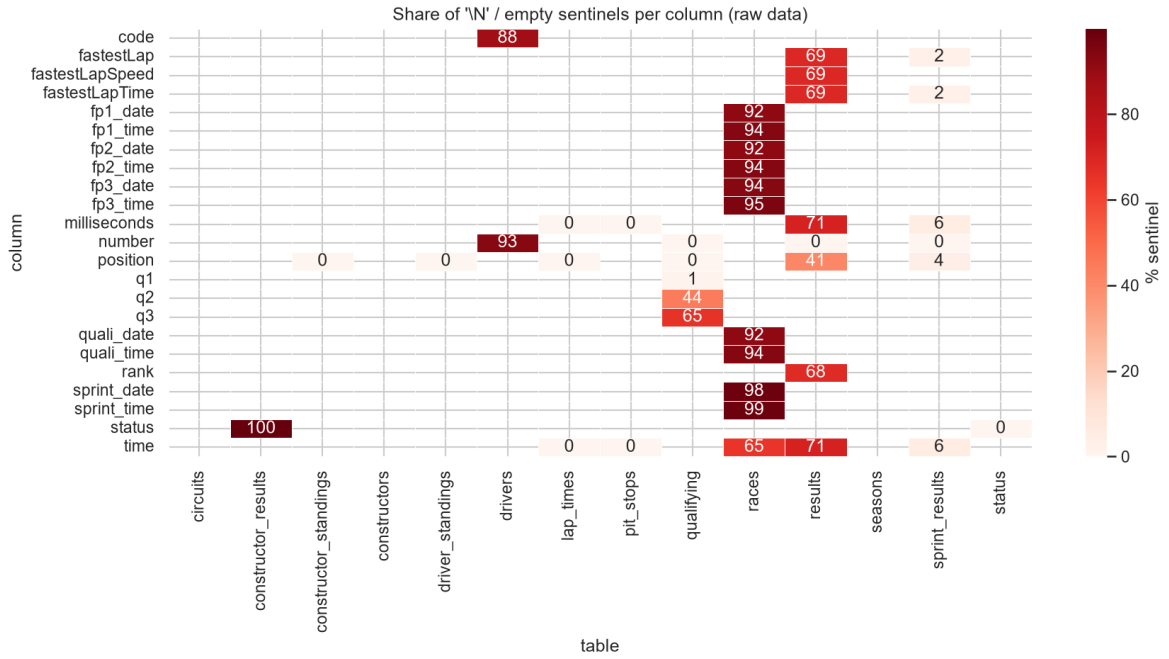


Figure 1 — Share of '\N' / empty sentinels per column (raw data).

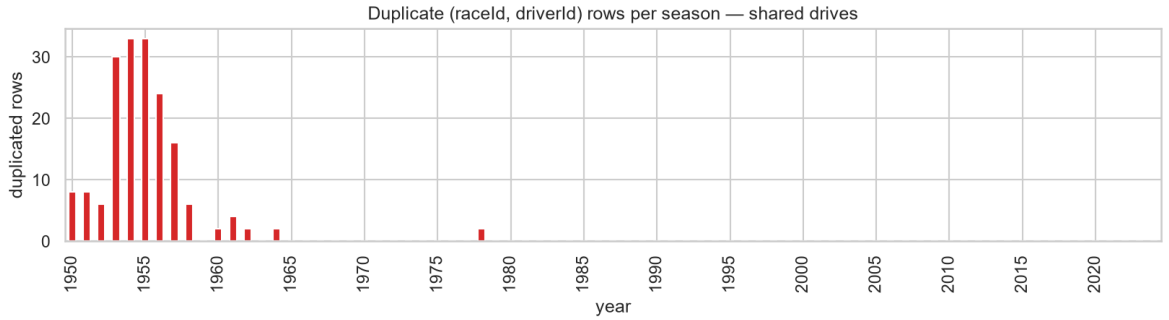


Figure 2 — Duplicate (raceId, driverId) rows per season: almost all in the 1950s (shared drives).

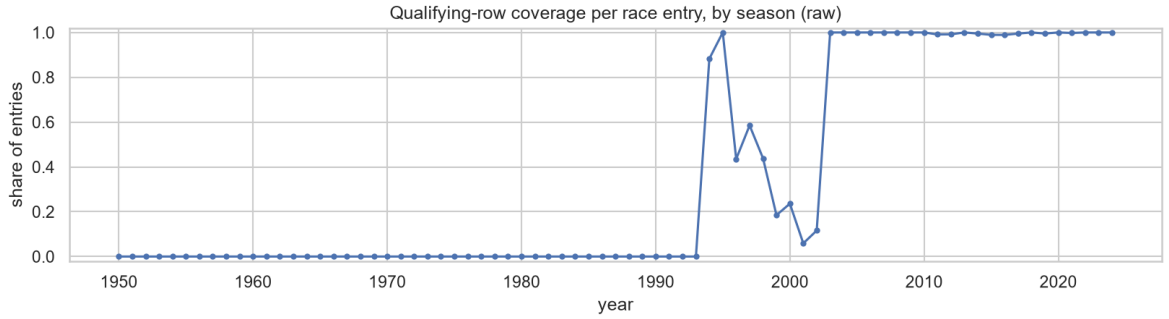


Figure 3 — Share of race entries with a qualifying row, per season (raw data). Coverage is complete only from 2003, which motivates the 2006+ modelling scope.

2.2 Cleaning pipeline and decisions

Cleaning works on copies of the raw tables; a hash check confirms the raw data is unchanged at the end. Each rule writes a row to *outputs/cleaning_log.csv*, and key, grain and foreign-key assertions run after every step that changes rows. Cleaning runs on **all seasons**, so the history features of a 2006 race can still use 2005. The modelling scope only selects rows afterwards. Table 1 lists the decisions.

#	Decision	Rule and justification	Effect
F	Sentinels, typing	strip, HTML-unescape, sentinels → NaN; IDs Int64, dates, durations → seconds. Missing values after typing must equal sentinels before (asserted).	0 parse failures
D1	Seasons	Model on 2006–2024 (knockout qualifying, complete data); 1950–2024 kept as a comparison trial (Section 7).	7,905 model rows
D2	Non-starters	Drop entries with no grid slot (DNQ, DNPQ, 107% rule; withdrew/excluded with grid 0). Keep 'Withdrew' with a grid slot: they were prediction rows at the cutoff.	−1,535 rows (9 in 2006+)
D3	Shared drives	Merge to one row per (race, driver): pre-race fields from the car the driver started in (earliest resultId); podium if any of his cars made the podium; points summed.	125 → 61 rows
D4	Indianapolis 500	Excluded (1950–60): separate set of drivers, 33-car grid, distorts the home-race analysis.	11 races
D5	grid = 0	2006+: pit-lane start → field size + 1, plus a flag. Pre-2006: slot unknown → NaN, plus a flag. Left as 0, a pit-lane start would look better than pole.	77 / 45 rows
D6	Missing features	Domain fill where the true value is known (debutant = 0 starts; first visit to a circuit = NaN + flag), then the training-set median plus missing-indicator columns.	11,360 / 12,516 cells
D7	Round-1 standings	Use last season's final standings (renamed teams matched by lineage) plus an is_round_1 flag.	75 races
D8	Timing outliers	Hard limits only: pit stop outside [1.5 s, 120 s]; qualifying time > 2× session median. A robust z-score would also remove genuine slow stops (13%).	482 cells → NaN
D9	Lap cleaning	Remove lap 1, pit in/out laps and laps > 107% of the race median; pace = driver median ÷ best median. Used only from earlier races.	513,483 of 589,081 laps
D10	Target	podium = positionOrder ≤ 3 after D3; every modern race has exactly 3 podiums (asserted).	375 races checked
D11	Status codebook	All 139 status labels mapped to finished / lapped / mechanical / incident / non-start / disqualified / ambiguous.	57 mechanical, 30 ambiguous
D12	Constructor lineage	Renamed teams grouped into families (e.g. Minardi → Toro Rosso → AlphaTauri → RB), with year ranges for IDs reused by unrelated teams. Tested against plain IDs (Section 7.5).	37 rules, 11 families
D13	Standings	No raw points: championship rank and share of the leader's points, which are comparable across scoring systems.	—
D14	Sprints	Sprint finish (known before the race) as a feature, 2021+.	18 weekends
D15	Countries	Nationalities and circuit countries → ISO-3166 codes; zero unmapped (asserted).	43 + 35 values
D16	Abnormal races	race_shortened flag (laps and race time < 75% of normal, or half points). For error analysis only, never a feature.	7 races

Table 1 — Cleaning decisions. F = fixed steps. Full counts are in outputs/decision_summary.csv.

2.3 Effect of cleaning (EDA after cleaning)

- *results* goes from 26,759 to 24,755 rows: −405 rows from the Indy 500, −1,535 non-starters, −64 from merged shared drives (Figure 4). All key, grain and foreign-key checks pass after every step.
- Missing shares in the original columns barely change (results 22.8% → 21.9%). Cleaning converts sentinels into real missing values; it does not invent or fill values. Filling is done per feature (D6), with the median fitted on the training split only.
- Outlier removal targets data errors: the 481 removed pit stops include ~51-minute red-flag suspensions, and the one qualifying time removed is 16:42.640 (Figure 5).
- **Class imbalance:** 14.2% of rows in the modelling scope are podiums (train 13.9%, validation and test 15.0%). This drives the metric choice in Section 6.

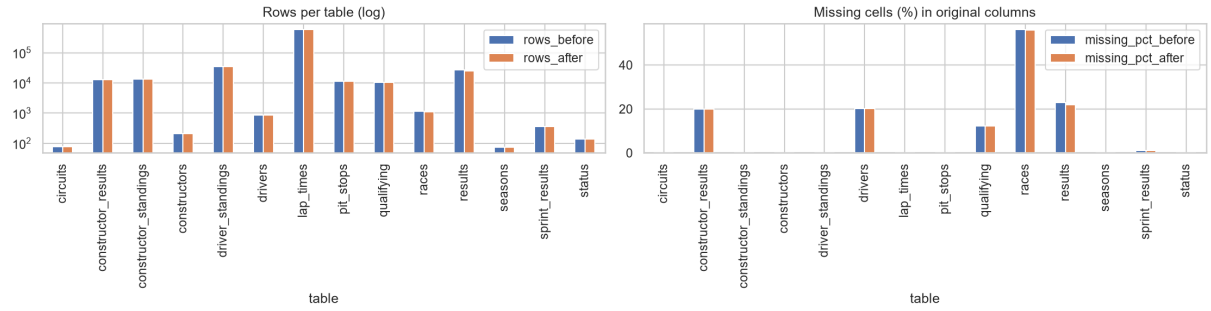


Figure 4 — Rows per table (log scale) and missing cells before vs after cleaning.

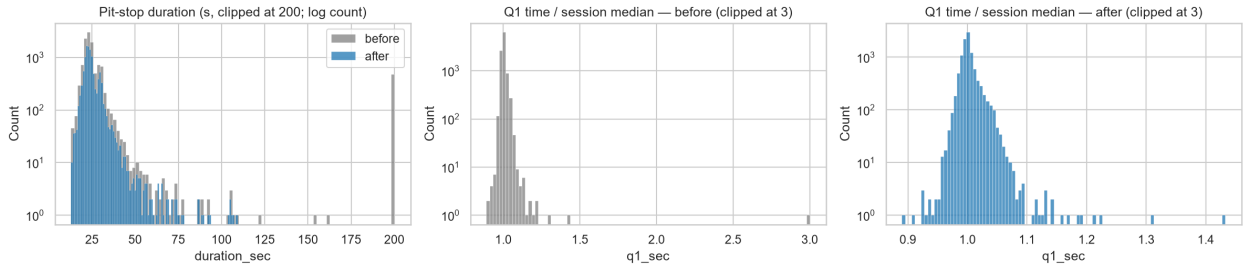


Figure 5 — Timing distributions before and after outlier handling (log counts). Left: pit-stop duration. Middle and right: Q1 time relative to the session median.

3. Data-engineering questions

Q1 — Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

- **Answer.** In 2014–2024 the best converters are Suzuka (89%, 16/18), Yas Marina (86%) and Monaco (85%). In 2003–2013 they are Magny-Cours (92%), Monaco and Monza (≈ 77%).
- **The ranking changes.** For the 15 circuits with at least 8 front-row starts in both eras, the Spearman rank correlation between eras is 0.21 ($p = 0.46$) using the grid and 0.50 ($p = 0.06$) using qualifying. Monza falls from joint first to eleventh; Yas Marina and Interlagos rise. Conversion increases almost everywhere after 2014: in the Mercedes and Red Bull years the front row was usually the fastest car.
- **Grid vs qualifying.** 74 entries (8.6% of qualifying front rows) differ between the two definitions, because penalties and pit-lane starts move drivers off the front row. Figures 6 and 7 show both rankings; the qualifying-based ranking is the more stable one across eras (0.50 vs 0.21).
- **Grain, missingness, cutoff.** One row per (race, driver), filtered to front-row starters. Only 2003–2024 is used, where qualifying is complete. Grid and qualifying position are both known before the start. With 8–22 starts per circuit and era, one podium moves a rate by 5–12 points.

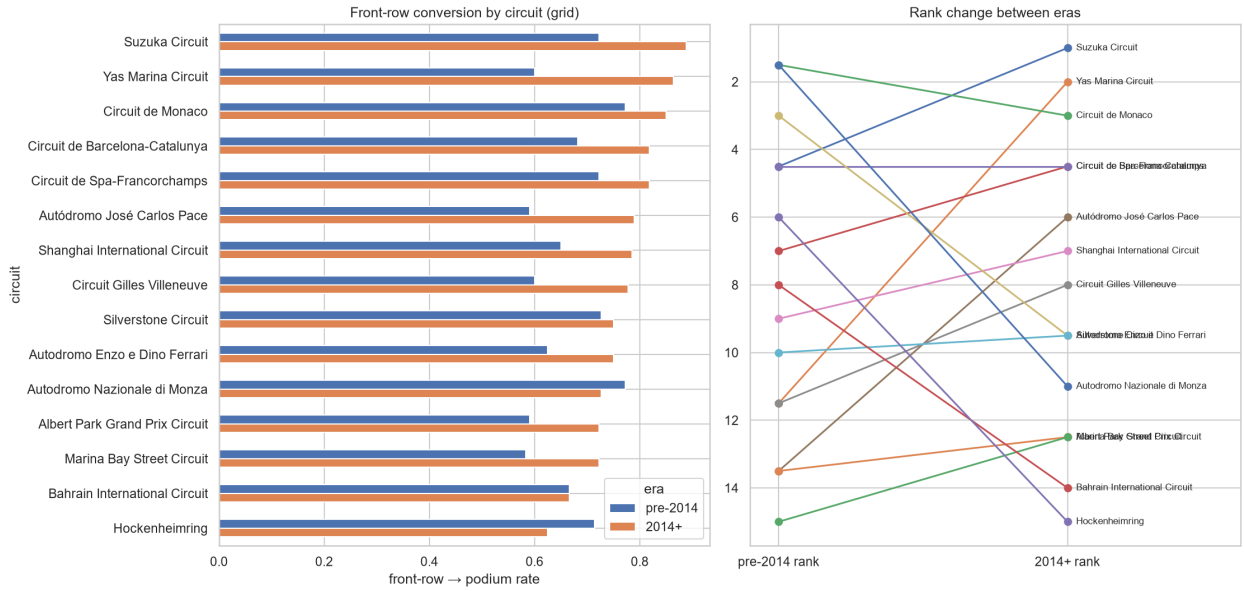


Figure 6 — Front-row conversion by circuit using the grid (after penalties), 2003–13 vs 2014–24, and the change in rank between eras.

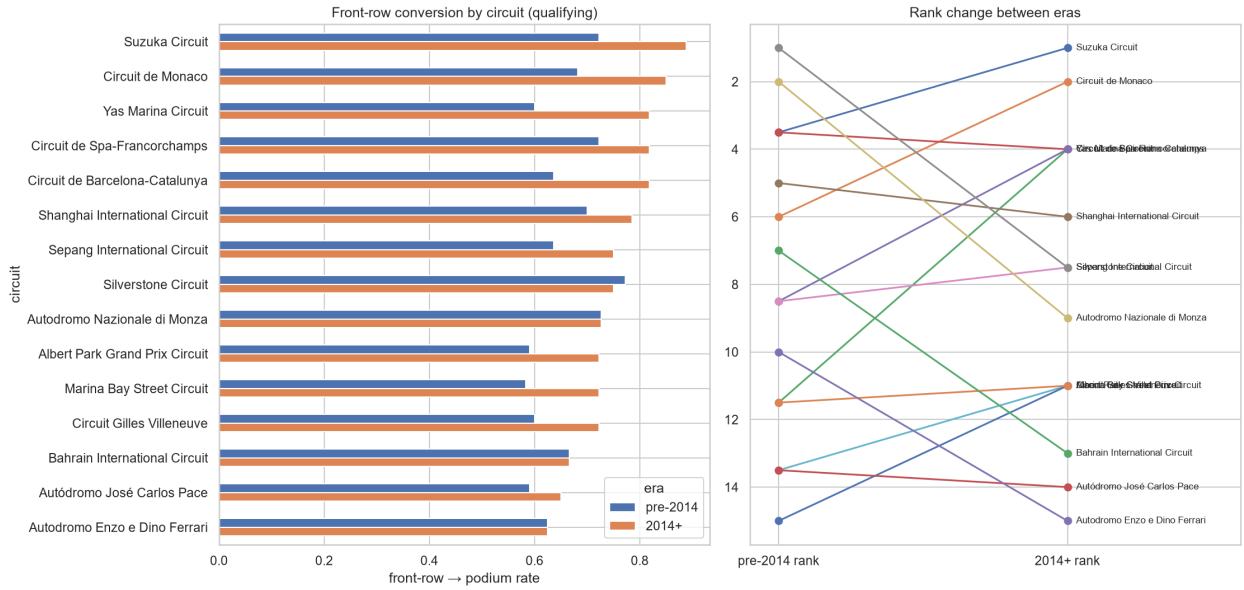


Figure 7 — The same using the qualifying position (before penalties).

Q2 — Do drivers reach the podium more often at home, controlling for starting position?

A home race means the driver's nationality and the circuit's country map to the same ISO code (D15). Drivers are compared within six grid strata (pole, P2–3, P4–6, P7–10, P11–15, P16+), and the strata are pooled with a Mantel–Haenszel odds ratio.

Period	Starters	Home	Crude OR	Grid-adjusted OR	95% CI
1950–2024 (main)	24,710	1,941	0.84	**0.95**	0.80–1.13
1950–2005	16,805	1,643	0.82	**0.95**	0.79–1.15
2006–2024	7,905	298	1.05	**0.87**	0.57–1.35

Table 2 — Home vs away podium odds. An OR above 1 would mean a home advantage.

- **Answer: no home advantage.** Home entries look worse overall (crude OR 0.84), but only because many home entries are one-off local drivers starting near the back. Within grid strata the effect disappears (OR 0.95, 95% CI 0.80–1.13), and the conclusion holds in every period.
- **Grain and missingness.** One row per starter with a known grid slot (24,710 rows, 1,941 home); zero unmapped nationalities. **Limitation:** nationality is not always where a driver grew up, and there are only 82

home entries on pole.

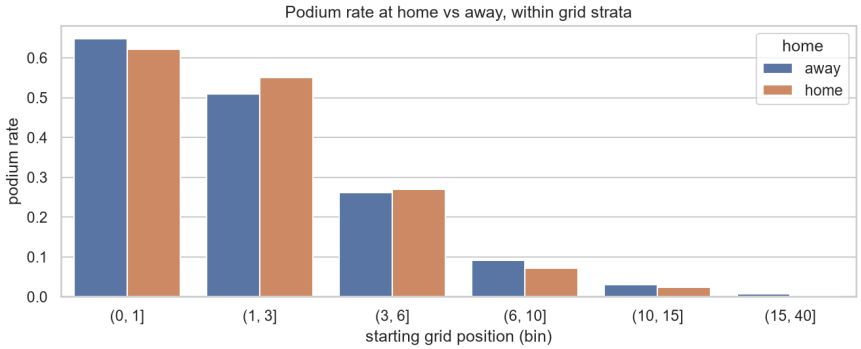


Figure 8 — Podium rate at home vs away within grid strata.

Q3 — Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

A mechanical retirement is a race start that ended because a car component failed. The status codebook (D11) defines it; ambiguous labels such as 'Retired', 'Puncture' and 'Wheel' are counted both ways: (a) not mechanical, (b) mechanical. The denominator is race starts, excluding drivers who withdrew after taking a grid slot. Table 3 groups renamed teams (D12) and labels each by its latest name in the era plus its family; Table 4 gives the same rates per constructor ID.

Era	Highest rates (variant a)	Lowest rate	Spearman (a) vs (b)
2014–2021	Caterham 23.5%, AlphaTauri (Faenza) 14.7%, McLaren 13.5%, Alpine F1 Team (Enstone) 13.5%	Mercedes (Brackley) 3.8%	0.95
2022–2024	Alpine F1 Team (Enstone) 11.0%, Sauber (Hinwil) 8.1%, Ferrari 5.9%, RB F1 Team (Faenza) 5.9%	Aston Martin (Silverstone) 3.0%	0.85

Table 3 — Mechanical-retirement rate per start, teams grouped by lineage (≥ 20 starts per era).

Era	Highest rates per constructor ID (variant a)
2014–2021	Caterham 23.5% (8/34), Lotus F1 17.3% (13/75), Toro Rosso 16.1% (39/242), McLaren 13.5% (43/318), Renault 13.5% (27/200)
2022–2024	Alfa Romeo 11.4% (10/88), Alpine F1 Team 11.0% (15/136), AlphaTauri 6.8% (6/88), Ferrari 5.9% (8/136), Williams 5.2% (7/134)

Table 4 — The same per constructorId (renamed teams split). The leaders are the same teams.

- **Answer.** 2014–21 is led by Caterham, Toro Rosso / AlphaTauri, McLaren (the McLaren-Honda years) and Lotus / Renault, with Mercedes the most reliable (3.8%). In 2022–24 Alpine and Alfa Romeo / Sauber lead, Aston Martin is the most reliable, and rates roughly halve across the field.
- **Robustness.** Counting ambiguous statuses as mechanical changes little: rank correlation 0.95 and 0.85, and the top team in each era is unchanged. Wilson 95% intervals (Figure 9) overlap for differences under about 4 points.

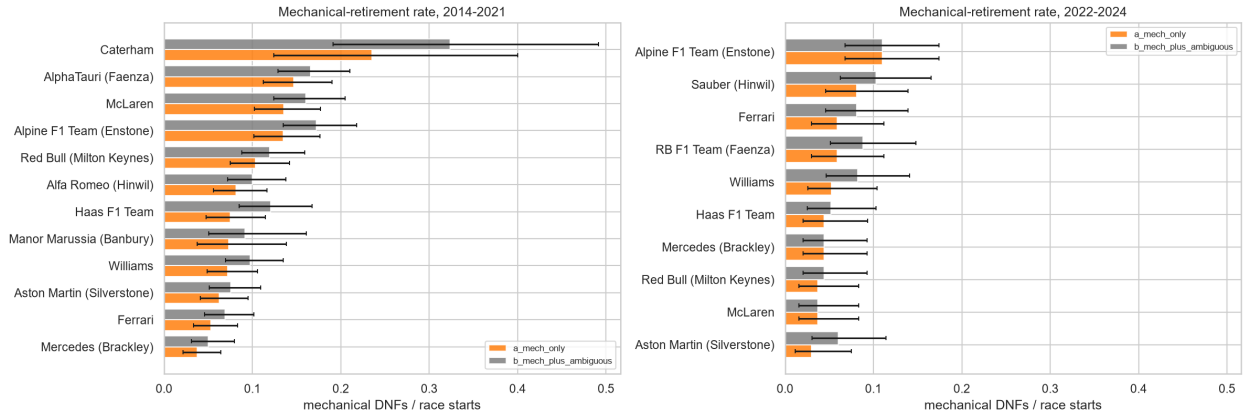


Figure 9 — Mechanical-retirement rate per team with 95% intervals; orange = mechanical only, grey = including ambiguous statuses.

4. Modelling table and features

The modelling table has exactly one row per (race, driver): 7,905 rows for 2006–2024 (asserted unique). Features are written to an explicit allowed list; the table may only contain listed features, IDs and the target. Table 5 lists the 26 selected features and why each is valid at the cutoff.

Group	Selected features	Valid at the cutoff because
grid	*pitlane_start*, *grid_minus_quali*, *field_size*	Known once the grid is fixed (qualifying + penalties).
qualifying	*quali_pos*, *gap_to_pole_pct*, *reached_q2*, *reached_q3*	Qualifying ends before the race.
standings	*drv_champ_rank*, *con_champ_rank*, *con_gap_to_leader_share*, *is_round_1*	Standings after the **previous** round; round 1 uses last season's final standings.
driver form	*drv_avg_finish_lastN*, *drv_podium_rate_lastN*, *drv_finish_rate_lastN*, *drv_pts_frac_lastN*	Rolling over the driver's previous 5 races (shift by one race).
constructor form	*con_podiums_lastN*, *con_pts_frac_lastN*	Rolling over the team family's previous 5 races, aggregated per race first.
pace pit	*drv_pace_ratio_lastN*, *con_pit_ratio_lastN*, *pit_data_available*	Clean-lap pace and pit-crew speed from **earlier** races only; pit data exists from 2011.
experience	*drv_career_starts*, *drv_career_podiums*, *drv_circuit_visits*, *drv_circuit_podium_rate*, *drv_circuit_first_visit*, *drv_age*	Counts and rates over earlier races; age at race date.

Table 5 — Selected features by group.

Leakage control

- Form, points-share, pace and pit features use only races dated before the race being predicted. An independent recomputation for 300 random rows matches the vectorised features exactly.
- A second, independent check: the inference function rebuilds all features from raw inputs using only races dated before the entry, and reproduces the modelling table for 300 random validation/test entries.
- Standings come from a race dated earlier, asserted for every row; round-1 standings come from the previous season. Raw championship points are never used (D13).
- Imputer and scaler are fitted on the training seasons only; the test set is scored once, after every choice was fixed on validation. The inference function rejects post-race fields such as *points*.

Feature selection

Selection uses the **training seasons only** (2006–2019), so validation and test never influence it. Rule 1 drops features that are empty or constant in training; rule 2 visits features by univariate AUC, highest first, and drops any whose |Spearman ρ| with an already-kept feature is ≥ 0.95. Of 36 candidates, 26 are kept. The strongest single features are *quali_pos* (univariate AUC 0.92), *gap_to_pole_pct* (0.90), *drv_pts_frac_lastN* (0.90) and *con_gap_to_leader_share* (0.90). Figure 10 shows the podium rate across the values of every candidate.

Dropped feature	Reason (training seasons 2006–2019)
grid	|ρ| = 0.968 with quali_pos (univariate AUC 0.918 vs 0.919)
grid_pct	|ρ| = 0.956 with quali_pos (univariate AUC 0.918 vs 0.919)
grid_unknown	constant in the training seasons (= 0)
gap_to_pole_sec	|ρ| = 0.988 with gap_to_pole_pct (univariate AUC 0.902 vs 0.903)
drv_pts_share	|ρ| = 0.958 with drv_champ_rank (univariate AUC 0.893 vs 0.895)
drv_gap_to_leader_share	|ρ| = 0.958 with drv_champ_rank (univariate AUC 0.893 vs 0.895)
con_pts_share	|ρ| = 1.000 with con_gap_to_leader_share (univariate AUC 0.897 vs 0.897)
con_best_finish_lastN	|ρ| = 0.971 with con_pts_frac_lastN (univariate AUC 0.891 vs 0.896)
is_sprint_weekend	constant in the training seasons (= 0)
sprint_finish_pct	empty in the training seasons

Table 6 — Dropped candidates. grid is not lost: it equals quali_pos + grid_minus_quali, both kept. The sprint features cannot be learned because sprints start in 2021 and training ends in 2019.

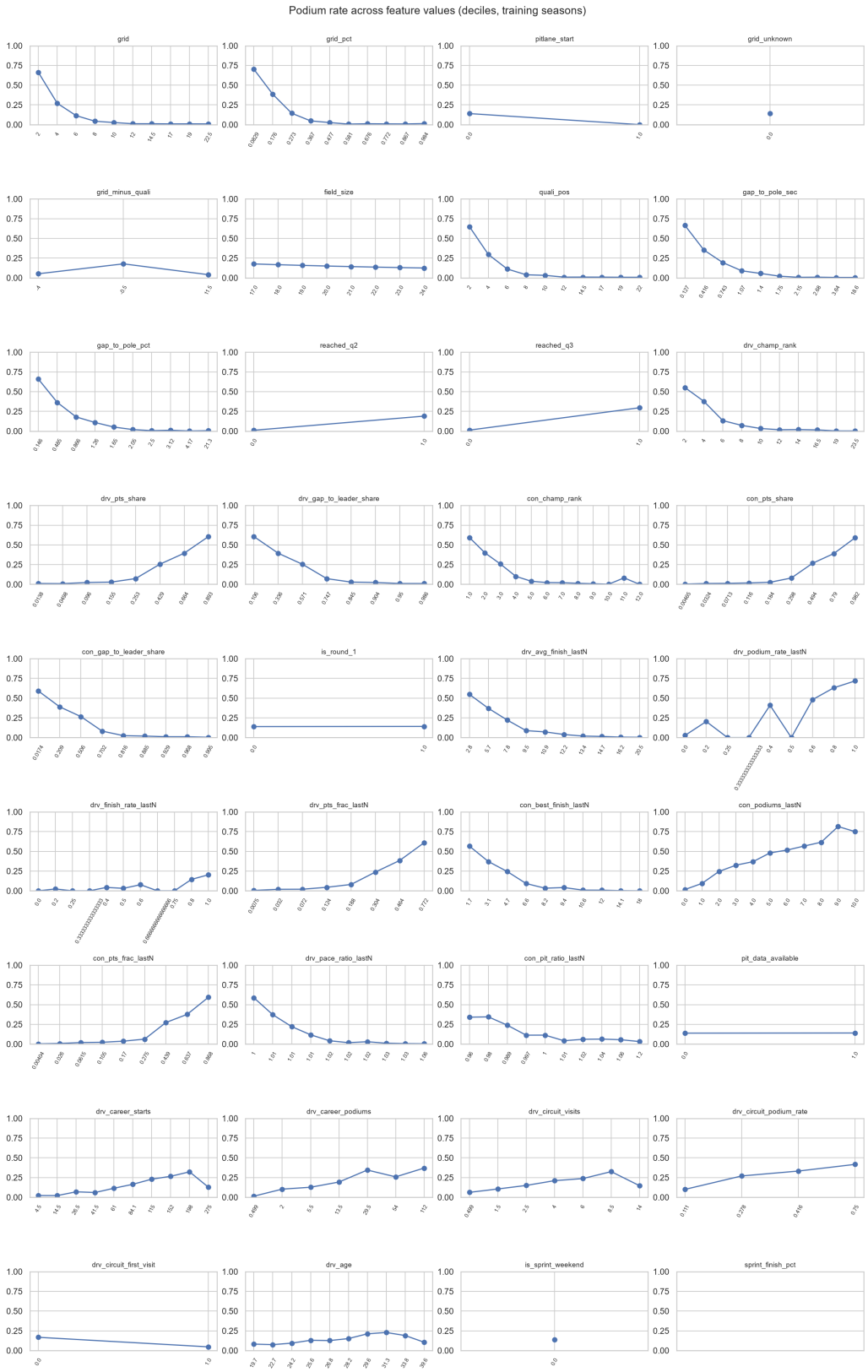


Figure 10 — Podium rate across feature values (deciles, training seasons): the evidence behind each feature.

5. Split and preprocessing

Rows are not independent: form carries over between races and the sport drifts from season to season. We therefore split **by season**, with the boundaries from the course's F1 guide: train 2006–2019 (5,768 rows, 268 races), validation 2020–21 (780 rows, 39 races), test 2022–24 (1,357 rows, 68 races). Podium rates are 13.9%, 15.0% and 15.0%. The test set is the whole ground-effect era (new car rules from 2022), so it measures how a model trained on the hybrid era carries over to a regulation change; validation never sees the

new era. The previous split (test 2023–24) is reported as a sensitivity check in Section 7.2.

Preprocessing is median imputation with missing-value indicators, followed by standardisation (tree models skip scaling); Figure 11 shows the effect on the distributions. Table 7 reports three controlled experiments; only the preprocessing changes, with the same logistic regression and features each time.

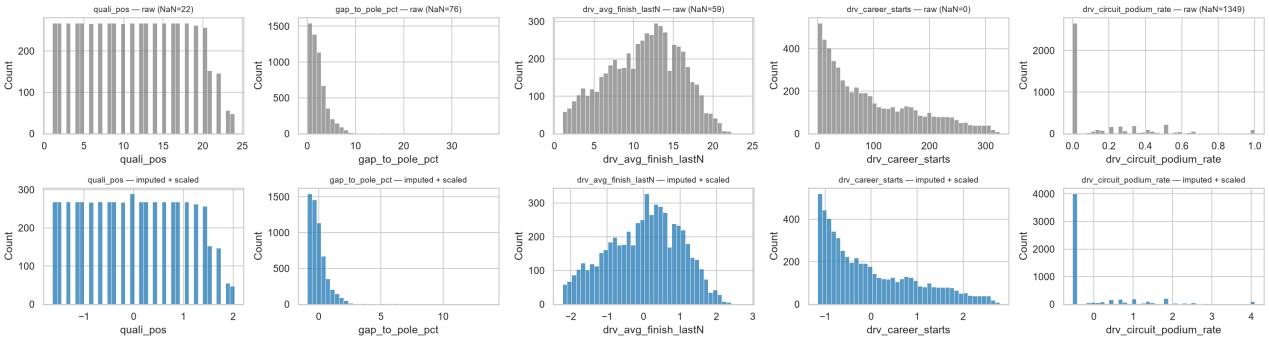


Figure 11 — Five features before (top) and after (bottom) imputation and scaling.

Experiment	Variant	Val PR-AUC	Val ROC-AUC	Val F1
E1 order	impute then scale	0.7454	0.9357	0.7137
E1 order	scale then impute	0.7422	0.9354	0.7137
E2 fit scope	fit on train only	0.7454	0.9357	0.7137
E2 fit scope	fit on all data (leaky)	0.7455	0.9358	0.7137
E3 indicators	add indicator=True	0.7454	0.9357	0.7137
E3 indicators	add indicator=False	0.7352	0.9341	0.7083

Table 7 — Preprocessing experiments (validation; F1 at the validation-chosen threshold).

- **E1, order:** impute → scale 0.7454 vs scale → impute 0.7422. Only about 2.7% of cells are imputed; scaling first places the imputed value at the median of the scaled column, so the difference is small.
- **E2, fit scope:** fitting the preprocessing on all data, including validation and test, brings no gain (0.7455 vs 0.7454) and would be leakage, so it is fitted on train only.
- **E3, missing indicators** help (+0.010 PR-AUC). Being missing is informative: no previous standings means a debutant or a new team, and no pit data means before 2011.

6. Models and evaluation

6.1 Models: how they work and their limitations

- **Reference rule** (not learned): score = 1 / grid position. Every model must beat it.
- **Logistic regression (baseline).** A weighted sum of the standardised features passed through a sigmoid, fitted by minimising the log-loss with an L2 penalty; class weights counter the 14% positive rate. Simple and interpretable, but only additive linear effects, and correlated features share their weights. Tuned over C.
- **Random forest.** Many decision trees on bootstrap samples with random feature subsets; their votes are averaged. Captures interactions without scaling, but deep trees memorise the training seasons. Tuned over depth × leaf size (best: depth 12, leaf 20).
- **Gradient boosting** (histogram-based). Shallow trees built one after another, each fitted to the errors of the previous ones with a small learning rate. Usually the strongest tabular model, but more hyperparameters and easy to overfit. Tuned over learning rate × depth × leaf (best: 0.03, depth 3, leaf 100).
- **Shallow neural network.** One or two ReLU hidden layers and a sigmoid output, trained with Adam on class-weighted cross-entropy, with dropout, L2 and early stopping on validation PR-AUC. Starting architecture 32-16 with dropout 0.2; six architectures tried; best: (64,), dropout 0.4 (2,881 weights). Limitations: stochastic, needs scaling, hard to interpret.

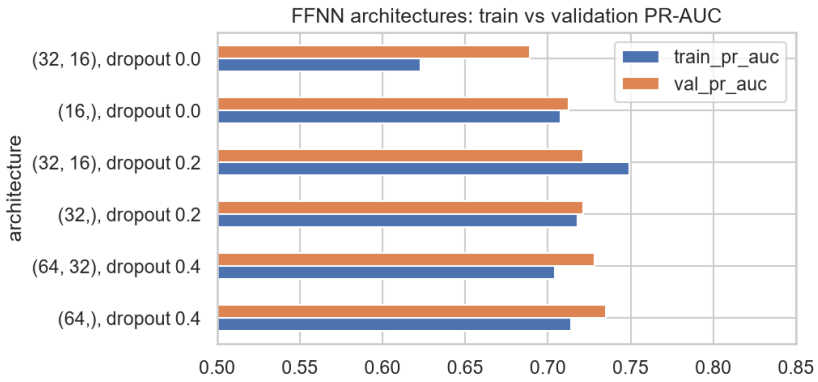


Figure 12 — Neural-network architectures: train vs validation PR-AUC.

6.2 Results on training, validation and test

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	PR-AUC train
Grid rule (reference)	0.909	0.639	0.692	0.692	0.692	0.645
Logistic regression, C = 1 (final)	**0.936**	**0.745**	**0.714**	**0.694**	**0.735**	**0.711**
Logistic regression, tuned	0.936	0.746	0.714	0.694	0.735	0.710
Random forest, default	0.923	0.689	0.714	0.703	0.727	1.000
Random forest, tuned	0.929	0.729	0.716	0.722	0.709	0.776
Gradient boosting, tuned	0.933	0.739	0.710	0.719	0.701	0.755
Shallow NN, starting architecture	0.924	0.721	0.714	0.703	0.727	0.749
Shallow NN, tuned	0.929	0.735	0.714	0.703	0.727	0.714

Table 8 — All five metrics on the validation seasons (2020–21); same rows and the same 26 features for every model. F1, precision and recall use the threshold chosen on validation.

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	PR-AUC train
Grid rule (reference)	0.914	0.647	0.649	0.650	0.647	0.645
Logistic regression, C = 1 (final)	**0.934**	**0.698**	**0.690**	**0.599**	**0.814**	**0.711**
Logistic regression, tuned	0.934	0.699	0.690	0.599	0.814	0.710
Random forest, default	0.926	0.642	0.653	0.604	0.711	1.000
Random forest, tuned	0.932	0.682	0.642	0.647	0.637	0.776
Gradient boosting, tuned	0.930	0.693	0.654	0.650	0.657	0.755
Shallow NN, starting architecture	0.930	0.683	0.648	0.630	0.667	0.749
Shallow NN, tuned	0.928	0.679	0.652	0.634	0.672	0.714

Table 9 — All five metrics on the unseen test seasons (2022–24); same rows and the same 26 features for every model. F1, precision and recall use the threshold chosen on validation.

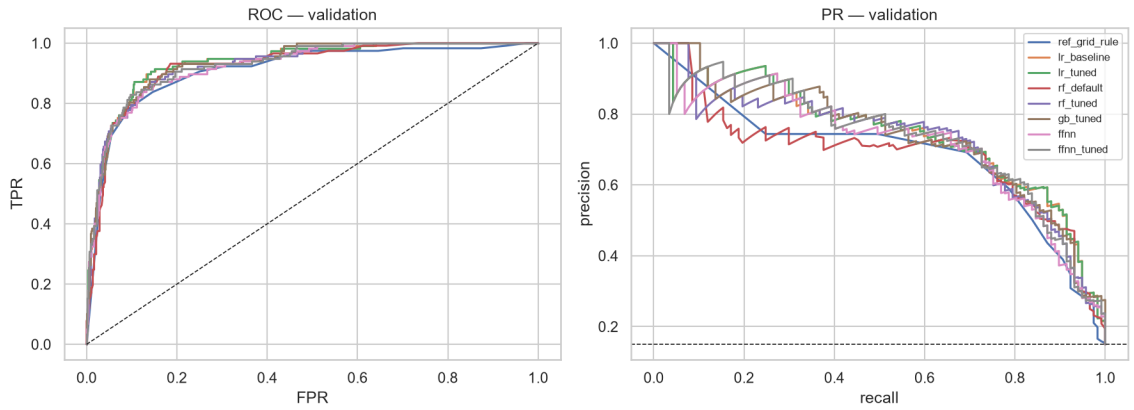


Figure 13 — ROC and precision–recall curves on validation.

6.3 Metric choice, threshold and model selection

- **Primary metric: PR-AUC.** Podiums are a minority (14–15%), and the task is ranking podium finishers above everyone else. ROC-AUC is inflated by the many easy back-of-grid entries: it is ≈ 0.93 for every model and 0.91 even for the grid rule, so it barely separates the models.
- **Secondary metric: F1** at a probability threshold chosen on validation: the quality of the actual yes/no prediction. The F1-maximising threshold was at least as good as 'exactly 3 podiums per race' for every model (logistic regression: 0.714 vs 0.709).
- **Model selection** (validation only): the simplest model within 0.005 of the best validation PR-AUC. Logistic regression with C = 1 (0.745) ties the tuned C = 10 (0.746) and beats every more complex model, so it is the final model.

6.4 Is the difference real? Race-bootstrap intervals

Validation and test hold only 39 and 68 races, so we resample whole races 1,000 times and compute 95% intervals, including paired intervals for the difference to the final model.

Model	Val PR-AUC [95% CI]	Δ vs final (val)	Test PR-AUC [95% CI]	Δ vs final (test)
Grid rule (reference)	0.639 [0.537, 0.755]	-0.107 [-0.158, -0.051]	0.647 [0.591, 0.709]	-0.051 [-0.095, -0.003]
Logistic regression, C = 1	0.745 [0.636, 0.850]	—	0.698 [0.637, 0.763]	—
Logistic regression, tuned	0.746 [0.636, 0.850]	+0.000 [-0.001, +0.002]	0.698 [0.638, 0.763]	+0.000 [-0.004, +0.003]
Random forest, default	0.689 [0.581, 0.801]	-0.056 [-0.109, -0.004]	0.642 [0.585, 0.714]	-0.057 [-0.093, -0.011]
Random forest, tuned	0.729 [0.623, 0.838]	-0.016 [-0.048, +0.013]	0.682 [0.624, 0.750]	-0.016 [-0.043, +0.011]
Gradient boosting, tuned	0.739 [0.637, 0.845]	-0.007 [-0.049, +0.033]	0.693 [0.639, 0.752]	-0.005 [-0.032, +0.020]
Shallow NN, starting architecture	0.721 [0.612, 0.826]	-0.024 [-0.049, +0.000]	0.683 [0.626, 0.743]	-0.015 [-0.037, +0.007]
Shallow NN, tuned	0.735 [0.621, 0.842]	-0.010 [-0.031, +0.012]	0.680 [0.623, 0.741]	-0.019 [-0.036, +0.002]

Table 10 — PR-AUC with 95% race-bootstrap intervals. A Δ interval that excludes 0 means a real difference from the final model.

- Only the grid rule (-0.051 on test, interval [-0.095, -0.003]) and the default random forest are clearly worse than the final model. Tuned forest, boosting and both networks are statistically indistinguishable from it, so choosing the simplest model of the tied group is justified.
- **Test result.** The final model reaches PR-AUC 0.698 on 2022–24, +0.051 over the grid rule, finding 166 of 204 podiums with 111 false alarms (precision 0.599, recall 0.814).
- **Across the rule change.** Validation → test falls from 0.745 to 0.698. By season: 2022 0.660 (first ground-effect season), 2023 0.770 (one dominant team), 2024 0.664 (four winning teams). ROC-AUC stays at 0.93–0.95: the model still separates the field, but ordering the front-runners gets harder.

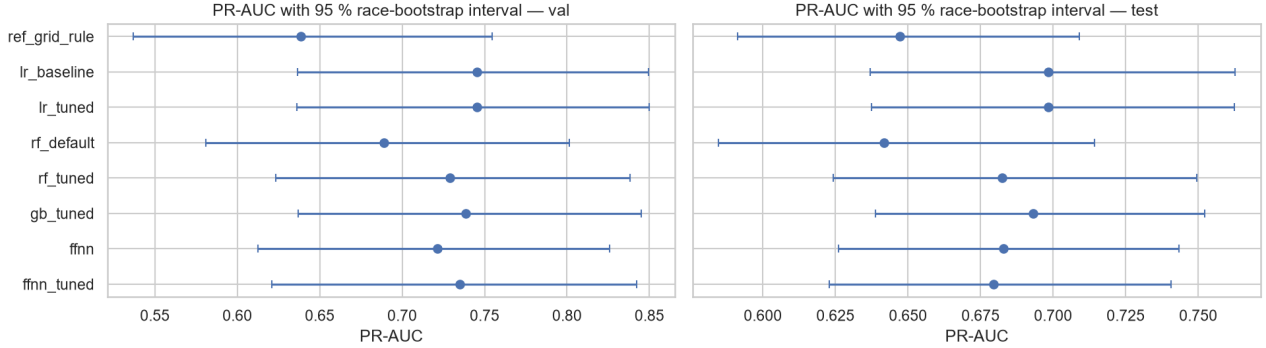


Figure 14 — PR-AUC with 95% race-bootstrap intervals.

6.5 Class weighting and calibration

Variant (validation)	PR-AUC	Brier score	Mean predicted	Distinct values
LR balanced (used)	0.746	0.0951	0.277	779
LR unweighted	0.741	0.0658	0.144	774
LR balanced + Platt	0.722	0.0669	0.150	775
LR balanced + isotonic	0.698	0.0673	0.151	64

Table 11 — Class weighting and calibration (observed podium rate 0.150). Calibration is scored with 5-fold cross-fitting by race on validation.

Class weighting barely changes the ranking but makes the raw scores over-confident (mean prediction 0.28 for a 15% rate). We keep the weighted model for ranking and thresholding and calibrate its score with **Platt scaling** fitted on validation: it is smooth and rank-preserving, while isotonic regression collapses the scores to 64 values on 780 rows. On test the Brier score falls from 0.112 (raw) to 0.069 (calibrated); Figure 15, right, shows the calibration curves.

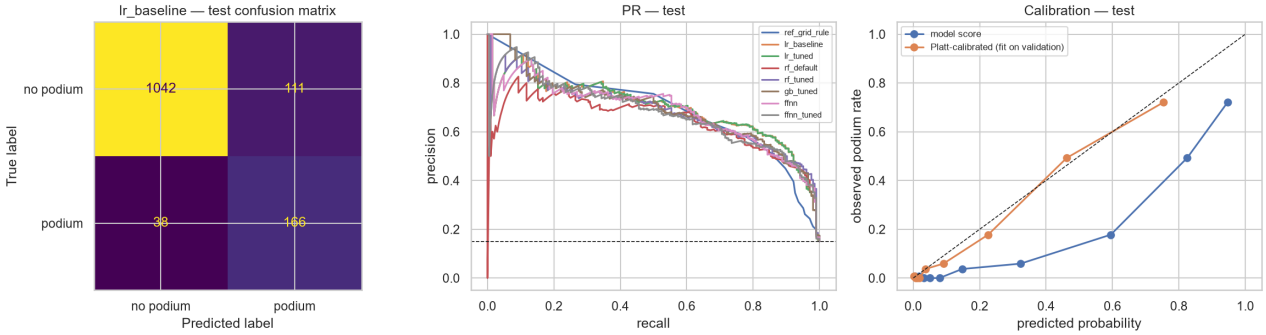


Figure 15 — Test set: confusion matrix of the final model, precision–recall curves of all models (dashed = base rate), and calibration of the raw and Platt-calibrated scores.

7. Experiments

7.1 Modern seasons vs all seasons (decision D1)

The same features and settings were trained on 2006–2019 and on 1950–2019 and scored on **identical** validation and test rows.

Training window	Train rows	Missing cells	LR val	RF val	NN val	LR test	RF test	NN test
modern (2006–2019)	5,768	2.7%	0.745	0.729	0.721	0.698	0.682	0.683
all (1950–2019)	22,618	21.1%	0.727	0.719	0.729	0.682	0.676	0.691

Table 12 — PR-AUC by training window (same validation and test rows).

Modern training is better for logistic regression and the random forest on both validation and test; the neural network is slightly better with all seasons (+0.008). The older data has 21% missing feature cells (no qualifying before 1994, unknown grid slots, no lap data before 1996) and different podium patterns (30+ car fields, high retirement rates). All gaps are within the bootstrap noise, so four times as many rows buys no measurable accuracy; the modern scope is kept for consistent definitions and the better result of the final model family.

7.2 Split sensitivity

Split	Model	Train rows	Val PR-AUC	Test PR-AUC	Test ROC-AUC	Test F1
Course guide (main)	Logistic regression, C = 1	5,768	0.745	0.698	0.934	0.690
Course guide (main)	Grid rule (reference)	5,768	0.639	0.647	0.914	0.649
Previous	Logistic regression, C = 1	6,108	0.722	0.713	0.928	0.679
Previous	Grid rule (reference)	6,108	0.629	0.664	0.911	0.667

Table 13 — The final model and the grid rule under the course split (train ≤ 2019, validation 2020–21, test 2022–24) and the previous split (train ≤ 2020, validation 2021–22, test 2023–24).

The model beats the grid rule by +0.05 PR-AUC under both splits. The course split's test score is lower because its test set starts with 2022, the least predictable season; the previous split had put 2022 into validation.

7.3 Feature-group ablation

Each group is removed in turn and the model re-trained with fixed hyperparameters; the threshold is re-chosen on validation. Done for the final model (95% paired race-bootstrap interval) and the tuned random forest (mean of 5 seeds ± seed std). One row removes grid and qualifying together, because grid was dropped as a near-duplicate of the qualifying position.

Removed group	LR val PR-AUC	LR Δ val [95% CI]	LR Δ test	RF Δ val (± std)	RF Δ test
none (full model)	0.745	—	—	RF full = 0.729	—
grid	0.722	-0.024 [-0.065, +0.007]	-0.010	-0.011 ± 0.005	-0.005
qualifying	0.649	-0.097 [-0.136, -0.043]	-0.104	-0.068 ± 0.004	-0.100
standings	0.733	-0.012 [-0.027, +0.003]	+0.005	+0.011 ± 0.004	-0.003
driver form	0.738	-0.008 [-0.020, +0.009]	+0.006	-0.010 ± 0.007	-0.000
constructor form	0.743	-0.002 [-0.011, +0.007]	+0.003	+0.007 ± 0.005	-0.003
pace pit	0.737	-0.009 [-0.020, +0.003]	+0.002	-0.003 ± 0.005	-0.003
experience	0.745	-0.000 [-0.019, +0.019]	+0.008	+0.014 ± 0.005	+0.007
grid + qualifying (starting position)	0.631	-0.114 [-0.153, -0.058]	-0.109	-0.093 ± 0.004	-0.104

Table 14 — Ablation (negative Δ = the group helps).

Only qualifying and the combined starting position have intervals that exclude 0: removing both costs 0.114 validation PR-AUC (0.109 on test). Grid, standings, form, pace and experience each add little on their own, and for the forest removing standings or experience even helps slightly. Grid, standings and form all have univariate AUC ≈ 0.89–0.92 (Section 4), but most of that signal overlaps with the starting position: univariate evidence measures signal, ablation measures *unique* signal.

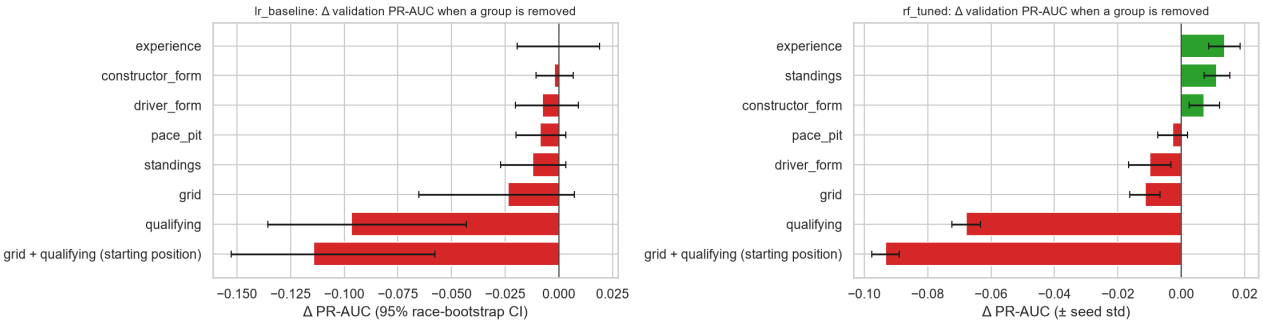


Figure 16 — Δ validation PR-AUC when a group is removed: final model (left, 95% race-bootstrap CI) and tuned random forest (right, ± seed std).

7.4 Form window

Window (races)	Val PR-AUC	Δ vs 5 [95% CI]	Test PR-AUC
3	0.733	-0.013 [-0.022, -0.002]	0.709
5 (chosen)	0.745	—	0.698
10	0.740	-0.006 [-0.029, +0.014]	0.701

Table 15 — Every rolling feature rebuilt with N = 3, 5, 10; final model re-trained.

Three races is measurably worse on validation (one DNF moves the average too much); ten is not distinguishable from five. Five is kept, now with evidence.

7.5 Constructor lineage vs plain constructorId (decision D12)

Constructor history	Val PR-AUC	Test PR-AUC	Train rows w/o standings	Val rows w/o standings
lineage families (D12, chosen)	0.745	0.698	0.5%	0.0%
plain constructorId	0.744	0.698	1.1%	0.8%

Table 16 — Constructor features by lineage family vs by plain ID. Δ val = -0.001, 95% CI [-0.008, +0.005].

The course guide suggests following *constructorId* and accepting that a rename resets history. The experiment shows lineage does not change accuracy measurably. We keep the families because they do not reset a real team's history at a rename (e.g. round 1 of 2020 AlphaTauri and 2021 Aston Martin and Alpine would otherwise have no standings) and because Q3 compares the same factory across eras. It is a documented judgement call, not a performance claim.

8. Explainability

These explanations are not causal. A feature being important means the model *relies* on it to predict, not that it *causes* podium finishes. Correlated features (e.g. qualifying position and reaching Q3) share and swap importance, and the sign of one feature can only be read given the others.

8.1 Global explanations

Permutation importance (drop in validation PR-AUC when a feature is shuffled) is dominated by *quali_pos* (0.477), *reached_q2* (0.069), *drv_career_podiums* (0.049), *gap_to_pole_pct* (0.027), *drv_pace_ratio_lastN* (0.024), *grid_minus_quali* (0.020). SHAP agrees (Figure 17): *quali_pos* dominates, followed by reaching Q2/Q3, recent pace, recent finishes, age and gap to pole.

- **Directions.** A better (lower) qualifying position, a smaller gap to pole, faster recent pace, better recent finishes and more career podiums push towards a podium. A pit-lane start and a grid penalty push strongly down.
- **Counter-intuitive signs.** *reached_q2* and *drv_age* push down. They are strongly correlated with *quali_pos* and career podiums, and in a linear model correlated features offset each other's weights. This is a concrete example of why importance must not be read as a causal effect.

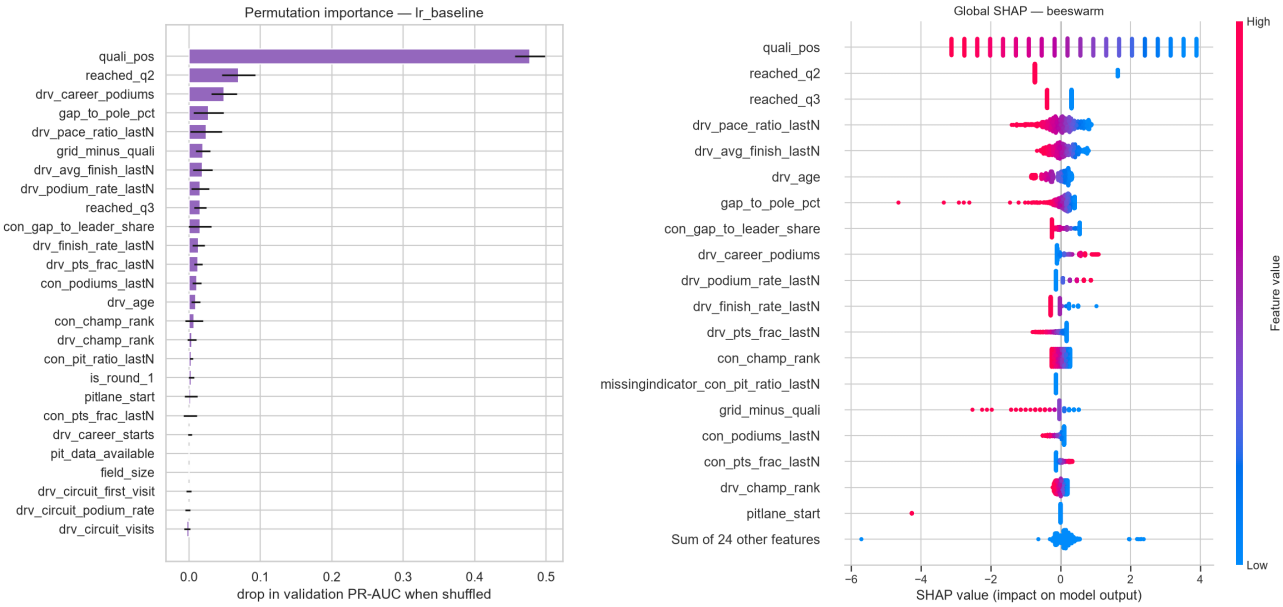


Figure 17 — Left: permutation importance on validation. Right: SHAP beeswarm in real units (colour = feature value; right = towards podium).

8.2 Local explanations

- **Confident hit:** Lewis Hamilton, 2021 Spanish Grand Prix, on pole. Predicted probability 0.98; he won. SHAP and LIME both put qualifying position first, then no pit-lane start, fast recent pace and good recent

finishes (Appendix B).

- **Missed podium:** Sebastian Vettel, 2020 Turkish Grand Prix, qualified 12th, started 11th. Predicted probability 0.04; he finished third in a wet, chaotic race. SHAP (Figure 18) shows the 6.9% gap to pole (qualifying was wet, which inflated the gaps) and *reached_q2* (the collinearity artefact) pulling down, and his 120 career podiums pushing up. A reasonable pre-race view: the outcome came from race conditions nobody could know at the cutoff.
- **LIME vs SHAP.** LIME runs on the raw features, so its rules read in real units ("11 < quali_pos ≤ 17", "gap_to_pole_pct > 3.03"), and it agrees with SHAP on the main drivers. Its local linear surrogate fits well for the confident hit (R^2 0.66) but poorly for the missed podium (R^2 0.29), where the probability is near 0 and the sigmoid is flat, so LIME is read qualitatively; SHAP is exact for a linear model.

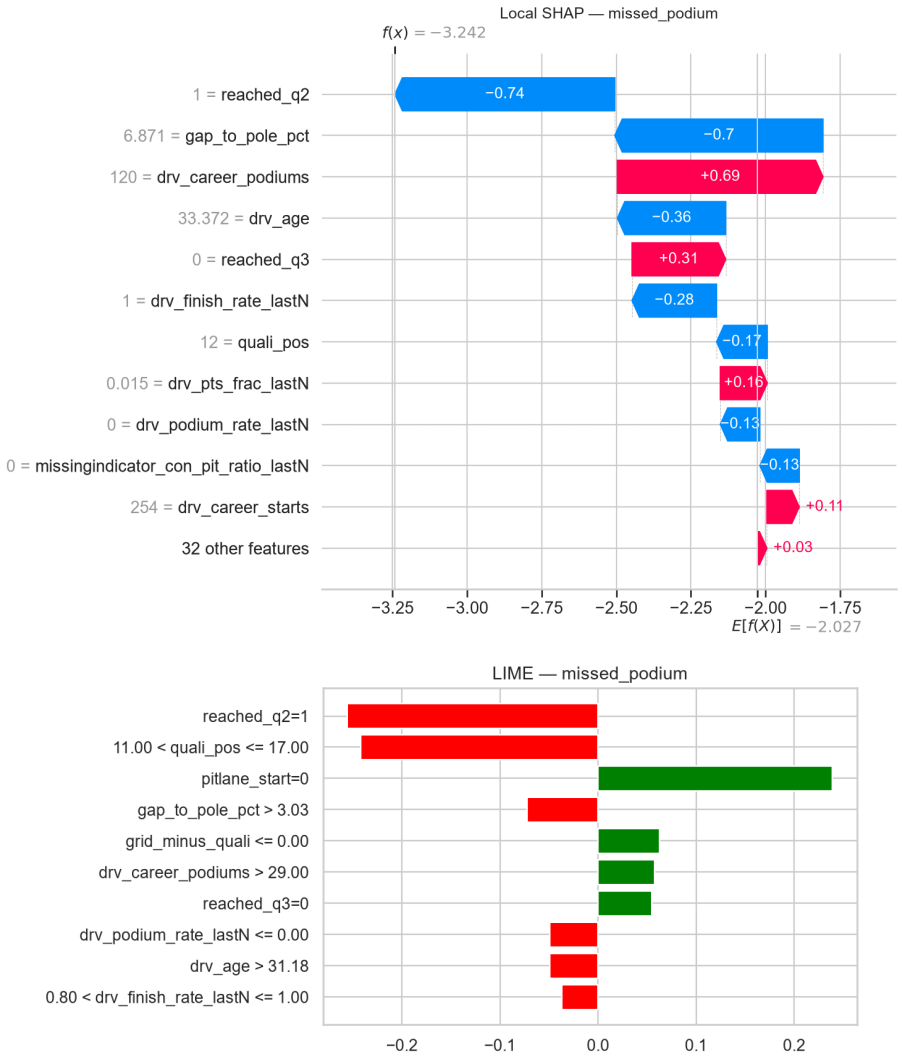


Figure 18 — Local explanations for Vettel, Turkey 2020. Top: SHAP waterfall (log-odds, real feature values). Bottom: LIME for the same row.

9. Inference function

predict_podium(entry) takes raw pre-race inputs for one driver in one race (driver, constructor, circuit, date, grid; optionally qualifying position, gap to pole, Q2/Q3 flags, field size, sprint result) and builds every feature itself from races dated before the entry. It returns the model score, the Platt-calibrated probability, the yes/no prediction at the validation threshold, and which features were filled by a fallback or imputed. Missing qualifying data is filled from the grid slot (with the typical gap for that position), because the pipeline's missing-value indicators would otherwise read 'not known' as 'set no time' and push the probability to ≈ 0. Missing history (a debutant) stays missing, exactly as for debutants in training. Post-race fields are rejected.

Record (Hülkenberg, Haas, Abu Dhabi 2024)	Score	Calibrated probability	Prediction	Filled / imputed
complete	0.296	0.080	no podium	0 / 0
no qualifying	0.152	0.037	no podium	4 / 0
debutant	0.327	0.090	no podium	3 / 8

Table 17 — Inference demo. Complete record: qualified 4th, started 7th after a grid penalty, finished off the podium. Without qualifying data the fallback assumes he qualified 7th, so the probability drops. Debutant: an unknown driver in the same car starting 5th.

10. Limitations

- **Qualifying-coverage drift.** Qualifying data is complete only from 2003, so the model is restricted to 2006+. Older seasons were tested (Section 7.1) and do not help.
- **Era sensitivity.** Regulation changes reshuffle team performance. The test set is a new era, and its first season is the hardest (PR-AUC 0.66 in 2022 vs 0.77 in 2023).
- **Small evaluation sets.** 39 validation and 68 test races; 95% intervals span about ±0.06–0.10 PR-AUC, so most model differences are not significant.
- **Calibration.** The class-weighted score is over-confident; reported probabilities are Platt-calibrated on validation.
- **Unused data.** Sprint features cannot be learned while training ends in 2019; pit-stop data starts only in 2011.
- **Small samples in the data questions.** 8–22 front-row starts per circuit and era (Q1); 82 home entries on pole (Q2).
- **Judgement calls.** Status grouping (D11), constructor lineage (D12, shown not to change accuracy) and 'nationality = home' are documented choices; Q3 is robust to the ambiguous statuses.
- **Unpredictable events.** First-lap incidents, safety cars and weather limit what any pre-race model can achieve (the Vettel example).
- **Inference fallback.** Filling a missing qualifying position from the grid hides a grid penalty.

11. Conclusion

A careful, fully asserted cleaning pipeline turns the relational Ergast tables into a leakage-safe modelling table with one row per driver and race. On the unseen 2022–24 seasons — a new regulation era — logistic regression reaches PR-AUC 0.698 against 0.647 for ranking by grid position, a gap that holds under both splits and whose bootstrap interval excludes zero. Forests, boosting and neural networks do not beat it, and the starting position carries most of the predictive signal. The data questions show that front-row conversion rankings changed in the hybrid era, that there is no measurable home advantage once grid position is accounted for, and that mechanical reliability improved markedly in 2022–24. Every decision, with the numbers behind it, is recorded in the notebook and in *DECISION_LOG.md*.

Appendix A — Judgement calls in the codebooks

Status codebook (D11, codebooks/status_map.csv). Illness, injury and similar → incident (driver-related, not a car failure). Underweight → disqualified (a regulation decision). Launch control and heat-shield fire → mechanical. Ambiguous (counted both ways in Q3): Retired, Technical, Not classified, Puncture, Tyre, Wheel, Wheel nut, Front/Rear/Broken wing, Damage, Debris, Out of fuel, Handling, Chassis, Stalled and a few rare labels.

Constructor lineage (D12, codebooks/constructor_lineage.csv). Families: Faenza (Minardi → Toro Rosso → AlphaTauri → RB); Silverstone (Jordan → MF1 → Spyker → Force India → Racing Point → Aston Martin); Hinwil (Sauber ↔ BMW Sauber, and Alfa Romeo 2019–23); Enstone (Toleman → Benetton → Renault → Lotus F1 → Renault → Alpine); Brackley (BAR → Honda → Brawn → Mercedes 2010+); Milton Keynes (Stewart → Jaguar → Red Bull); Banbury (Virgin → Marussia → Manor); Caterham (Lotus Racing → Caterham); Arrows (with Footwork); Ligier → Prost; March ↔ Leyton House. The IDs for Alfa Romeo, Mercedes and Renault are shared with unrelated works teams of the 1950s–80s, so these rules carry year ranges. Not linked: Tyrrell → BAR (new team, new factory).

Appendix B — Remaining figures from the notebook

Every figure the notebook produces is in this report; those not shown above follow, in notebook order.

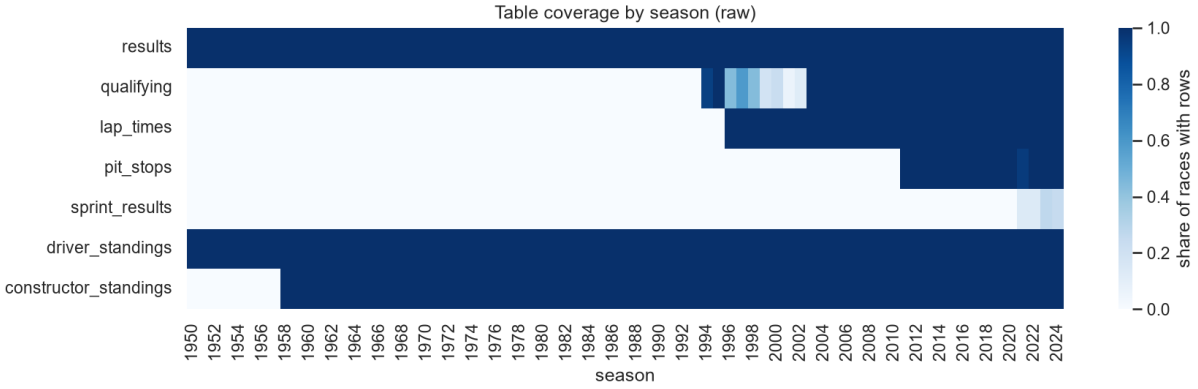


Figure B1 — Table coverage by season (raw): share of races with rows in each table.

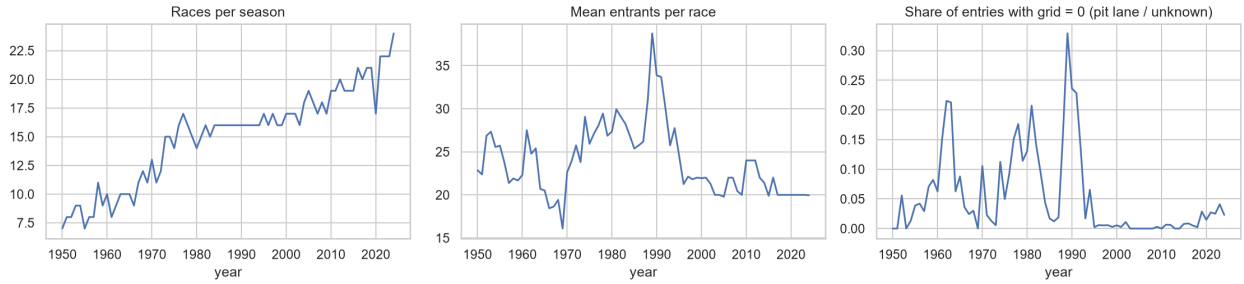


Figure B2 — Races per season, mean entrants per race, and share of entries with grid = 0.

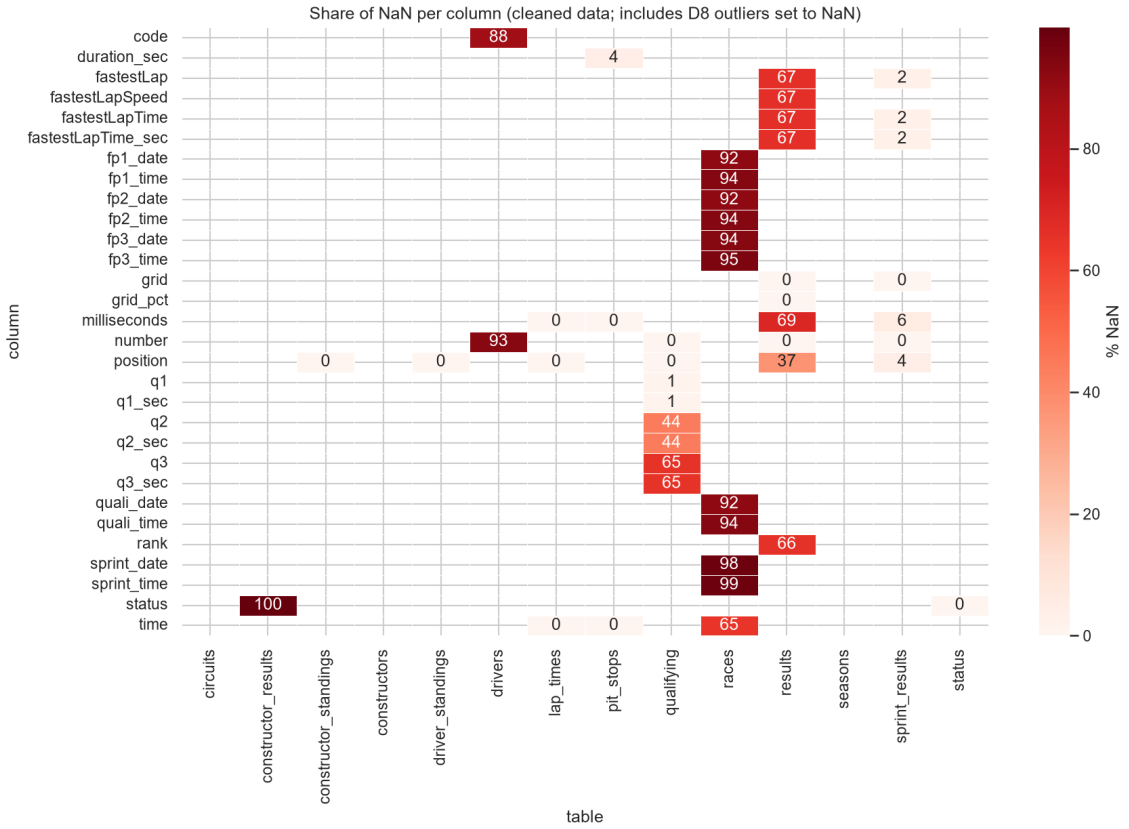


Figure B3 — Share of NaN per column after cleaning (compare with Figure 1).

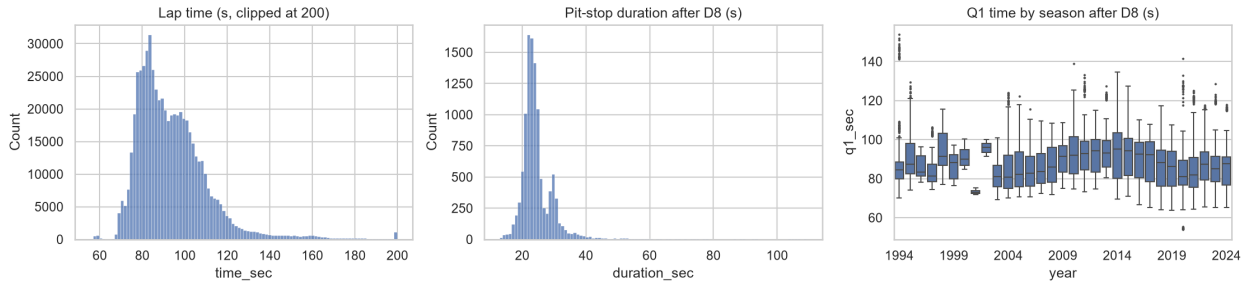


Figure B4 — Typed duration columns after cleaning: lap times, pit stops, Q1 by season.

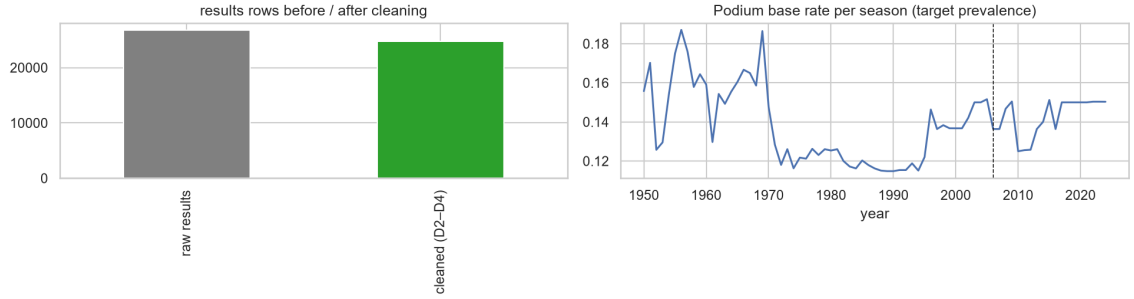


Figure B5 — Result rows before/after cleaning, and podium base rate per season.

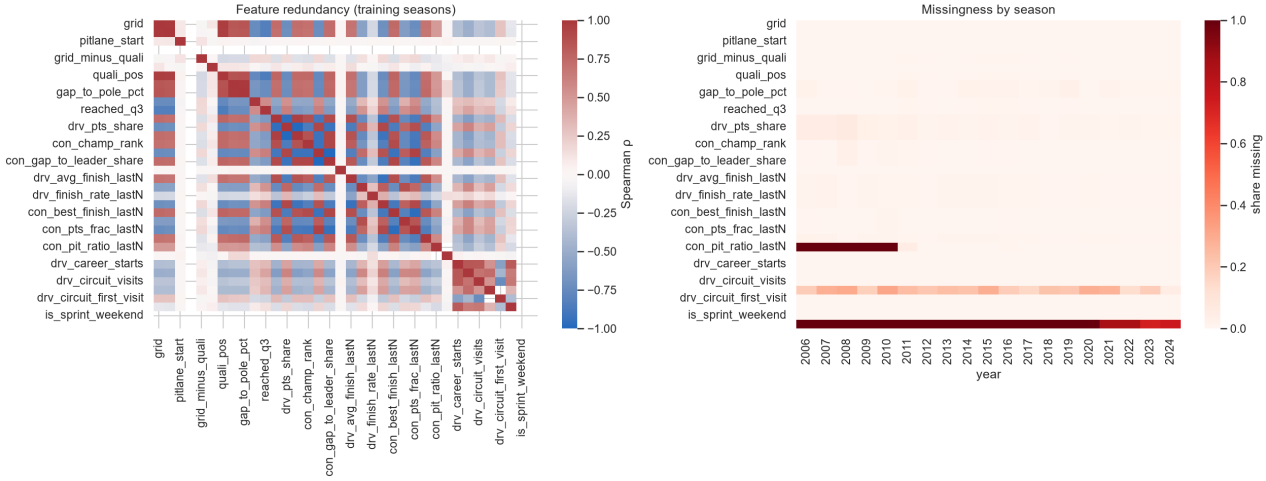


Figure B6 — Feature redundancy (Spearman ρ, training seasons) and missingness by season.

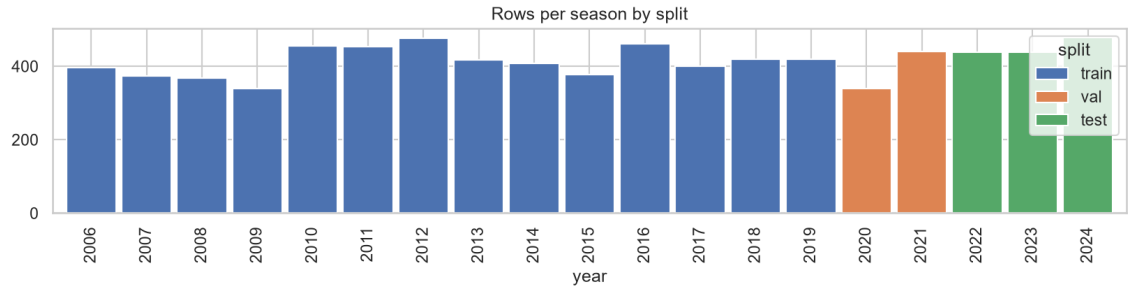


Figure B7 — Rows per season by split.



Figure B8 — Logistic-regression coefficients (standardised features).

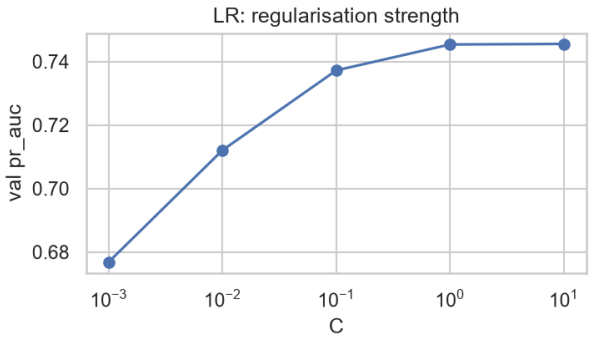


Figure B9 — Logistic regression: validation PR-AUC vs regularisation strength C.

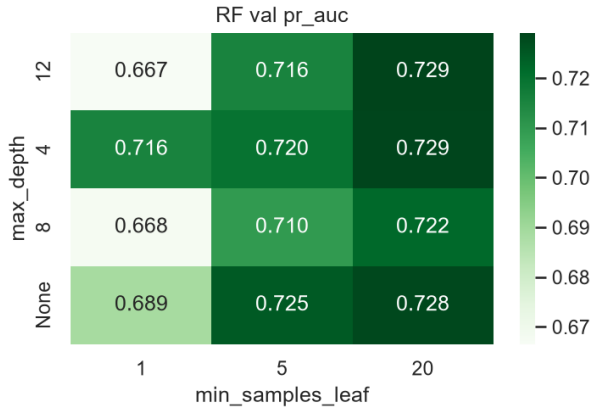


Figure B10 — Random forest: validation PR-AUC over depth × leaf size.

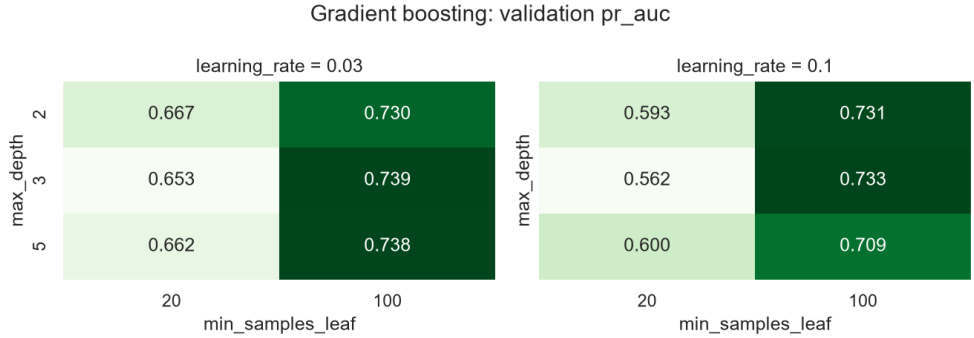


Figure B11 — Gradient boosting: validation PR-AUC over learning rate × depth × leaf size.

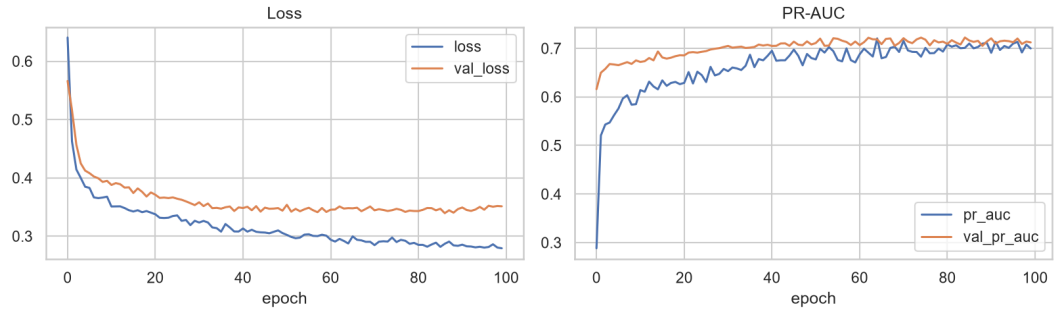


Figure B12 — Neural network (starting architecture): loss and PR-AUC per epoch.

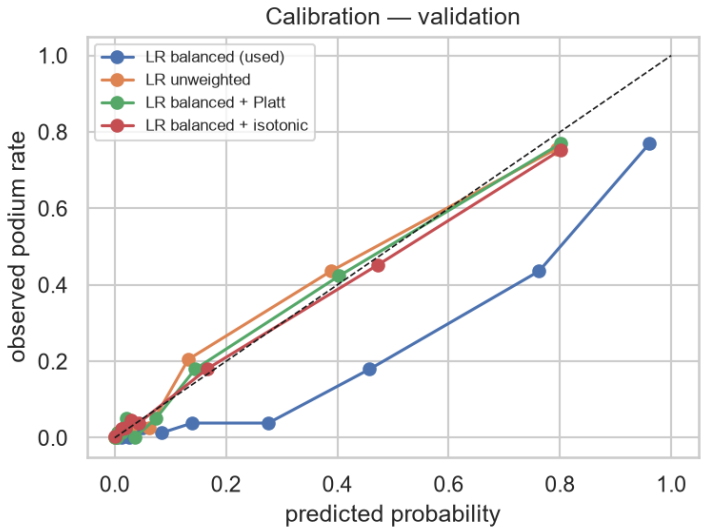


Figure B13 — Calibration on validation: weighted, unweighted, Platt and isotonic.

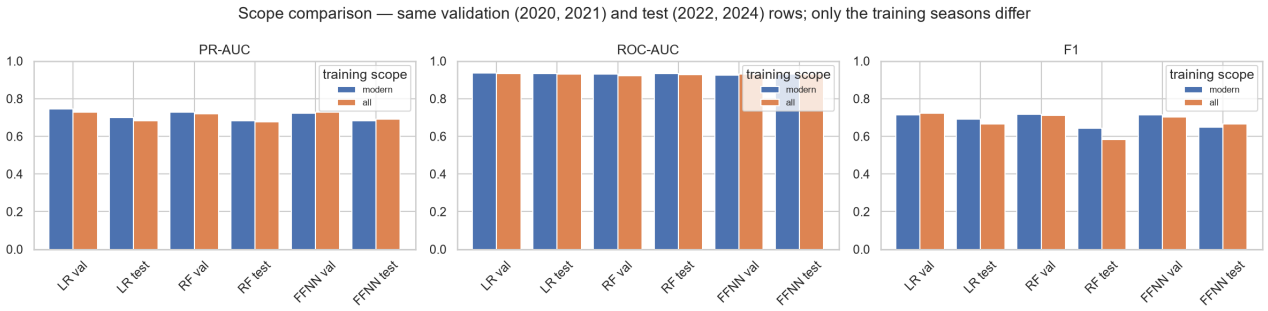


Figure B14 — Scope comparison on validation and test (Table 12).

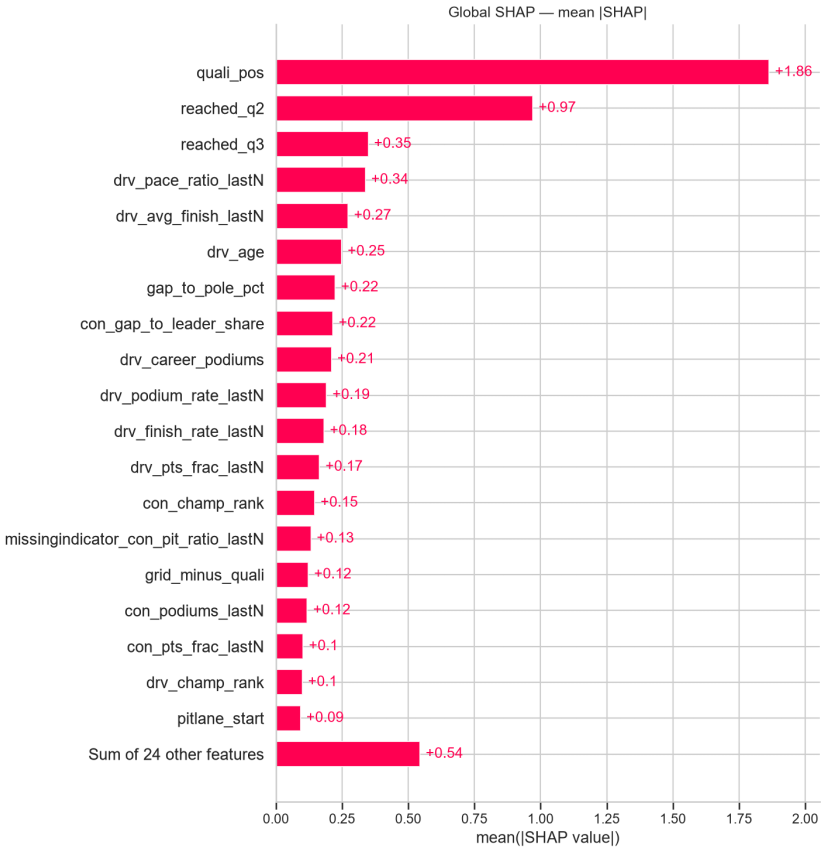


Figure B15 — Global SHAP: mean |SHAP| per feature.

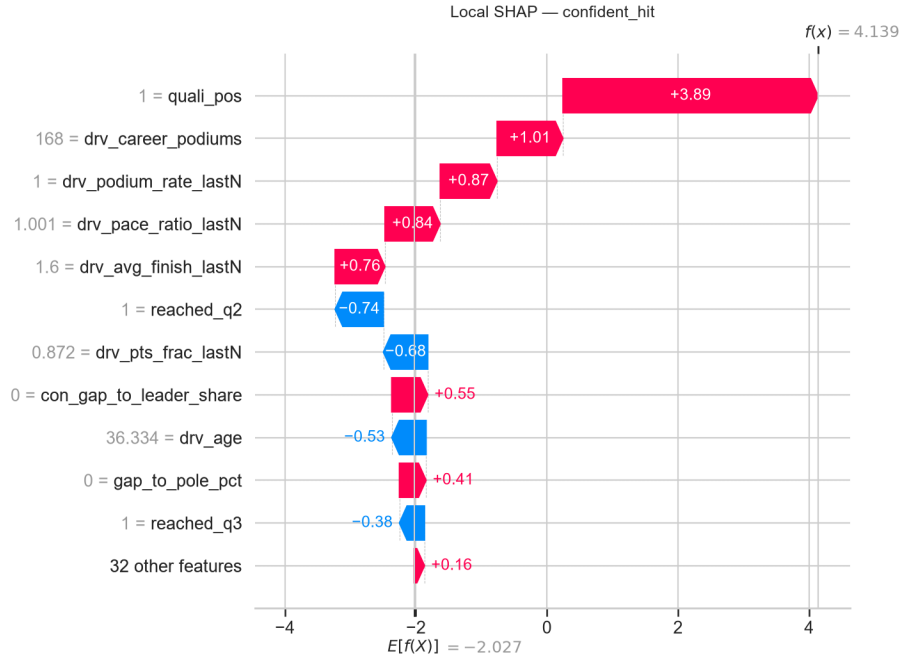


Figure B16 — Local SHAP for Hamilton, Spain 2021 (confident hit).

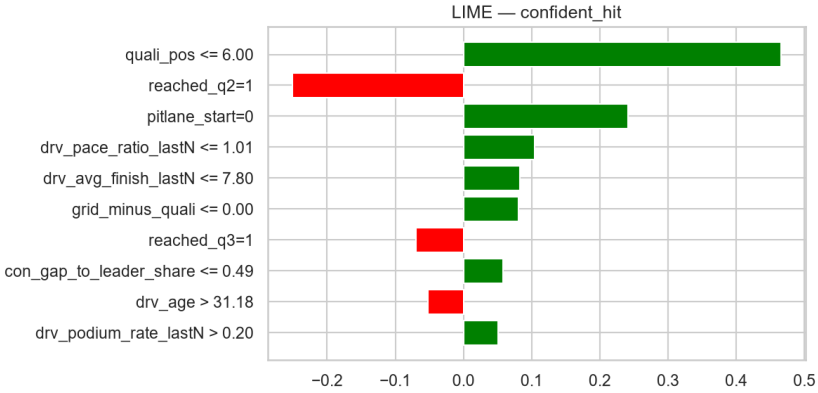


Figure B17 — LIME for Hamilton, Spain 2021 (confident hit).

Appendix C — Reproducibility

Code and outputs: github.com/omarkhashab/F1-Podium-Prediction. Install *requirements.txt*, run *F1_Podium_Prediction.ipynb* top to bottom (about 8 minutes on a laptop CPU), then *python report/build_report.py* to regenerate this report. Every table above is read from *outputs/*.csv* and every figure from *outputs/figures/*; a fresh re-run reproduces the output tables exactly.